Chapter 7 exercises

DS07-E01 • Stability and inversion count

Run stable insertion sort on `(5,A),(2,B),(4,C),(2,D)`. List each pass, the initial strict inversion count, and shift count. What fails if the movement condition becomes greater-than-or-equal?

DS07-E02 • Three-way partition and stack depth

Trace lt, i, gt on `[4,2,4,7,1,4]` with pivot 4 using the stated partition algorithm. Prove that recursing on the smaller side and looping on the larger limits stack depth to O(log n), but does not ensure O(n log n) worst-case time.

DS07-E03 • The model behind the lower bound

Prove the worst-case Ω(n log n) comparison-sorting bound for distinct keys using a decision tree. Explain why counting sort does not contradict it and why the n! argument cannot directly describe an all-equal-key input.

DS07-E04 • A counterexample to unstable LSD passes

Apply decimal LSD radix sort to `31,14,22,11`. Suppose the second pass reverses lower-digit order among equal tens digits. Give the incorrect output. Compare the costs of a very large base b and a very large digit count d.

DS07-E05 • Compare three partition contracts

Input [4,2,4,7,1,4]. Run the chapter's Hoare partition using the first key 4; list i, j and swaps. Then run Lomuto with last key 4 and ≤ on the left. Give returned boundaries and recursive ranges, and compare the three-way result.

DS07-E06 • Equal keys and termination

Run the chapter's Hoare partition on five equal keys, showing indices and returned j. What sizes result from last-pivot Lomuto with ≤ on the left? Why can three-way partition finish immediately?

DS07-E07 • Adjacent swaps and stability

Run bubble sort with early exit on [3,1,2,0], listing passes, initial inversions and swap count. Run ordinary selection sort on [2A,2B,1] and discuss stability.

DS07-E08 • Shell groups and stability

Run Shell sort on [8,5,3,7,6,2] using gaps=[3,1], showing each pass. Then use gaps=[2,1] on [5A,5B,2,8] and explain the final relative order of 5A and 5B.

DS07-E09 • Winner-tree merge

Four sorted runs are [2,9], [4,8], [1,7], [3,6]. Show the first three outputs, the candidate leaves after each replacement and the new winner. Why is the largest valid key not a general exhaustion sentinel?

DS07-E10 • Stable merge and comparison bound

Merge [2A,5] with [2B,4], listing comparisons, cursors and output. For two nonempty sorted runs of lengths m and n, what is the maximum number of key comparisons? Explain.

References and attribution

Lin Cing, handwritten data structures notes; MIT 6.006 sorting and comparison lower bounds; ISO C++ sorting requirements.

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https://github.com/xinhi6716/computer-science-self-study